

2023 - EE OHIKOA

B4

$$y = 2x^2 - 4x + 3$$

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$$y' = 4x - 4$$

$$4x - 4 = 0$$

$$x = 1 \rightarrow y = 1$$

$$E(1, 1)$$

Ardeatutia ebakite puntuok

$$OY \rightarrow x = 0 \rightarrow y = 3 \quad (0, 3)$$

$$OX \rightarrow 0 = 2x^2 - 4x + 3$$

$$x = \frac{4 \pm \sqrt{4^2 - 4 \cdot 2 \cdot 3}}{2} = \frac{4 \pm \sqrt{-4}}{2}$$

$$y = x^2 - 2x + 3$$

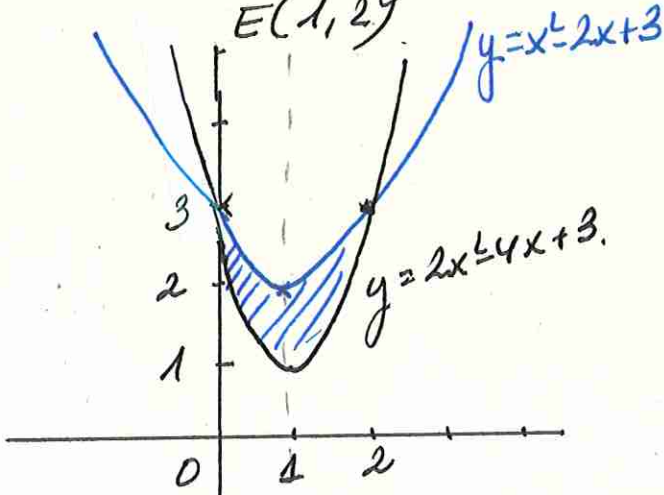
$$y' = 2x - 2$$

$$2x - 2 = 0$$

$$x = 1 \rightarrow y = 2$$

$$E(1, 2)$$

$$OY \Rightarrow x = 0 \rightarrow y = 3 \rightarrow (0, 3)$$



EBAKITE PUNTUAK

$$\begin{cases} y = 2x^2 - 4x + 3 \\ y = x^2 - 2x + 3 \end{cases}$$

$$2x^2 - 4x + 3 = x^2 - 2x + 3$$

$$x^2 - 2x = 0 \quad \begin{cases} x_1 = 0 \rightarrow y = 3 \\ x_2 = 2 \rightarrow y = 3 \end{cases}$$

$$A = \int_0^2 (x^2 - 2x + 3) - (2x^2 - 4x + 3) dx = \int_0^2 (-x^2 + 2x) dx$$

$$= \left[-\frac{x^3}{3} + x^2 \right]_0^2 = \left(-\frac{2^3}{3} + 2^2 \right) - \left(-\frac{0^3}{3} + 0^2 \right) =$$

$$= -\frac{8}{3} + 4 = \frac{-8 + 12}{3} = \frac{4}{3} \text{ u}^2$$